

Supplementary Information 1

A registered genome-wide analysis identifies robust, context-dependent transcriptional responses of lychee to *Peronophythora litchii*

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Contents

This file contains Figures S1–S3 and their captions. Supplementary Data 2, supplied as `Discover_Plants_Data_and_Code.zip`, contains machine-readable Tables S1–S18, all figure source data, the registered protocol and amendment log, analysis code, workflows, configurations, tests, metadata, and environment specifications. The complete archive is also permanently available at <https://doi.org/10.5281/zenodo.22436625>.

- Figure S1: replicate-level normalized counts.
- Figure S2: conditional parametric power curves.
- Figure S3: exploratory signed-signature estimate.

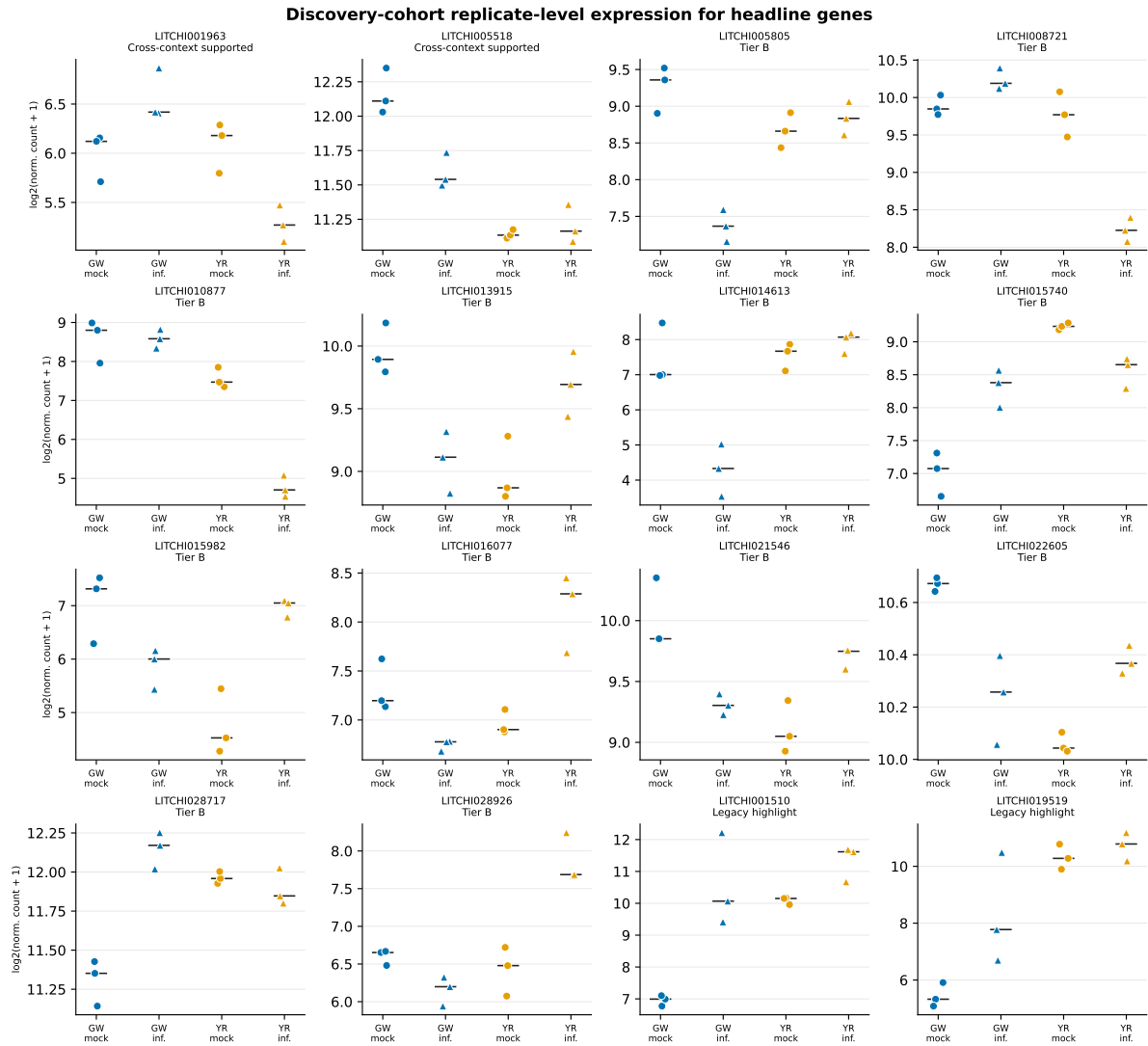
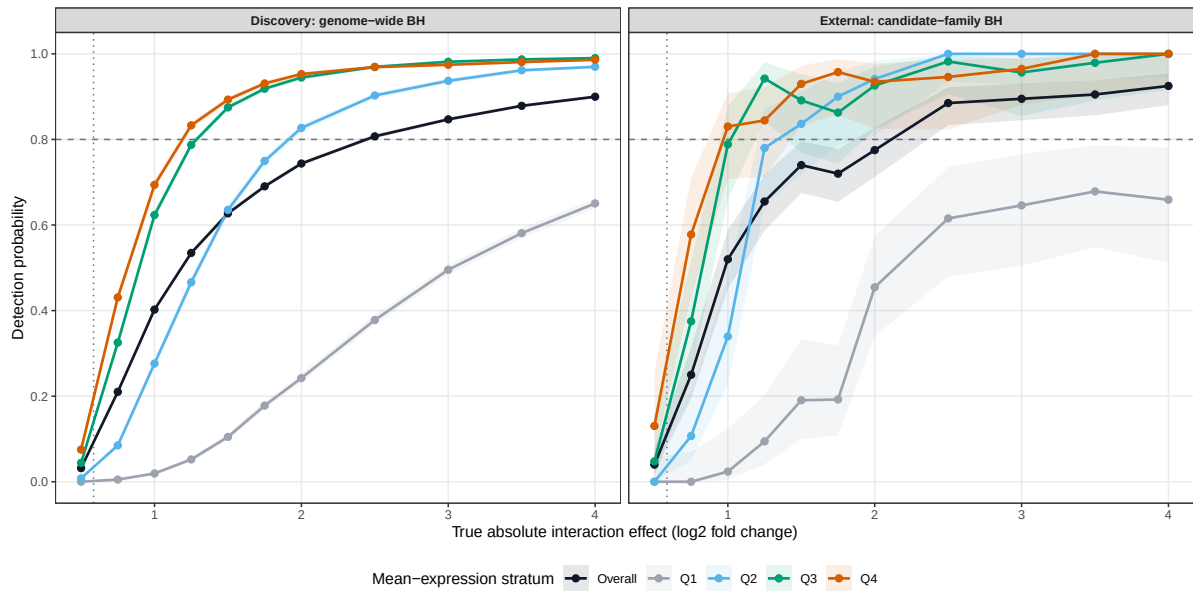


Figure S1: Replicate-level normalized counts for the two cross-context-supported genes, twelve Tier B genes, and two legacy highlights. Points are individual deposited libraries; bars show cultivar-treatment medians.



Dashed line: 80% power. Dotted line: registered $|\log_2\text{FC}|$ threshold. Parametric power is conditional on fitted means, dispersions, deposited-library independence, and the simulated non-null prevalence.

Figure S2: Parametric detection probability for cultivar-by-infection effects under genome-wide discovery and candidate-family external adjustment. Curves show the overall result and mean-expression quartiles; the dashed line marks 80% power.

S3 Exploratory signature estimate (not confirmatory)

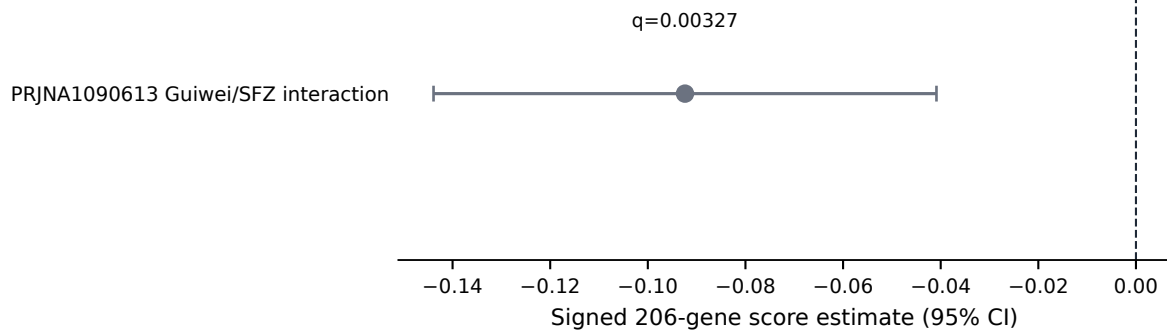


Figure S3: Exploratory PRJNA1090613 signed-signature estimate, shown separately because cohort metadata did not support its inclusion in the confirmatory evaluation.